

2520090088(B.DHARSHAN)

SKILL 11

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root@PC: ~/os_lab/history_pipeline_lab
root@PC:~# cd ~/os_lab
mkdir -p history_pipeline_lab && cd history_pipeline_lab
root@PC:~/os_lab/history_pipeline_lab# nano history_manager.c
root@PC:~/os_lab/history_pipeline_lab# gcc -g -O0 -Wall -pthread history_manager.c -o history_manager
./history_manager
=== Command History (Capacity: 5) ===
 1  ls -la
 2  pwd
 3  cd /tmp
 4  echo 'Hello World'
 5  export VAR=123
=====

[ACTION] Adding 6th and 7th commands (overwriting oldest entries)...
=== Command History (Capacity: 5) ===
 3  cd /tmp
 4  echo 'Hello World'
 5  export VAR=123
 6  gcc -Wall main.c -o main
 7  ./main
=====

[TEST] Retrieving entry #1 (should be overwritten/missing):
ID 1: NOT FOUND (Overwritten)

[TEST] Retrieving entry #6:
ID 6: gcc -Wall main.c -o main

root@PC:~/os_lab/history_pipeline_lab# nano pipeline_exec.c
root@PC:~/os_lab/history_pipeline_lab# gcc -g -O0 -Wall -pthread pipeline_exec.c -o pipeline_exec
./pipeline_exec
=== Pipeline Execution Test: echo | grep | wc ===

[EXEC] Running 3-stage pipeline...
Output (Count of lines containing 'a'):
2

[SUCCESS] Pipeline completed successfully.
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root@PC: ~/os_lab/history_pipeline_lab
[SUCCESS] Pipeline completed successfully.
root@PC:~/os_lab/history_pipeline_lab# valgrind --leak-check=full ./history_manager
valgrind --leak-check=full ./pipeline_exec
==562== Memcheck, a memory error detector
==562== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==562== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==562== Command: ./history_manager
==562==
=== Command History (Capacity: 5) ===
 1 ls -la
 2 pwd
 3 cd /tmp
 4 echo 'Hello World'
 5 export VAR=123
=====

[ACTION] Adding 6th and 7th commands (overwriting oldest entries)...
=== Command History (Capacity: 5) ===
 3 cd /tmp
 4 echo 'Hello World'
 5 export VAR=123
 6 gcc -Wall main.c -o main
 7 ./main
=====

[TEST] Retrieving entry #1 (should be overwritten/missing):
ID 1: NOT FOUND (Overwritten)

[TEST] Retrieving entry #6:
ID 6: gcc -Wall main.c -o main

==562==
==562== HEAP SUMMARY:
==562==    in use at exit: 0 bytes in 0 blocks
==562==   total heap usage: 1 allocs, 1 frees, 1,024 bytes allocated
==562==
==562== All heap blocks were freed -- no leaks are possible
==562==
==562== For lists of detected and suppressed errors, rerun with: -s
==562== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

```
root@PC: ~/os_lab/history_pipeline_lab
[TEST] Retrieving entry #1 (should be overwritten/missing):
ID 1: NOT FOUND (Overwritten)

[TEST] Retrieving entry #6:
ID 6: gcc -Wall main.c -o main

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==563== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==563== Command: ./pipeline_exec
==563==
=== Pipeline Execution Test: echo | grep | wc ===

[EXEC] Running 3-stage pipeline...
Output (Count of lines containing 'a'):
2

[SUCCESS] Pipeline completed successfully.
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==563== HEAP SUMMARY:
==563==    in use at exit: 0 bytes in 0 blocks
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==563== For lists of detected and suppressed errors, rerun with: -s
==563== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
root@PC:~/os_lab/history_pipeline_lab#
```